MIT-ADT UNIVERSITY
PUNE, INDIA
A Leap Towards World Class Education

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Subject: PL-II (Python) Roll No.: 37
Name of the Experiment: Assignment - 02 Exp. No.: 02

Performed on:
Submitted on:

Marks	Teacher's Signature with date
10	

1) List the python sequence data types and state one key characteristic of each with respect to mutability or ordering?

→ i) list: Ordered and muttable (elements can chaged)
ii) Tuple: Ordered and immutable (cannot be modified)
iii) String:- Ordered and immutable sequence of Character
iv) Set:- Unordered, mutable, store unique elements only.
v) Dictionary:- ordered (python 3.7+), mutable, Stores data as key value pairs.

2) Explain how indexing and slicing work in python sequence. Illustrate how they differ when applied to string and list?

→ Indexing- It is used to access a single element using its position (start from 0)
Slicing- It is used to access a part of the sequence

Incomplete for Theory ☐ Diagrams ☐ Observation Table ☐
Calculations ☐ Graphs ☐ Results ☐ Conclusion ☐
Understanding Through Q/A ☐ Late Submission ☐ Neatness ☐

Teacher's Signature

using [start : end : step]

Example :-

S = "Python"

I = [10, 20, 30, 40]

Indexing :

S[1] → 'y'

I[2] → 30

Slicing :

S[1:4] → "yth"

Difference :-

→ String cannot be modified after slicing (immutable)

→ List can be modified after slicing (mutable

3) Write a program that:

a) Uses indexing and slicing on a string list, and

b) Applies atleast two built-in function to apply display meaningful results.

→
```
text = "Himanshu"
print("First character:", text[0])
print("Substring:", text[0:3])
print("length of string:", len(text))
print("Uppercase:", text.upper())
```

Output:

First character: H ; Substring: Hima ; length of string: 7
; Uppercase: HIMANSHU

4. Analyze the difference betw equality (==) and identity (is) when used with sequence type. Discuss how object mutability and memory reference affect program behaviour?

→ Equality (==) checks whether values are same. Identity (is) checks whether both variable point to the same memory location.

Example:-

```
a=[1,2,3]
b=[1,2,3]
print(a==b)
print(a is b)
```

Output:

True
False

Explanation:-

- Lists are mutable and stored in different memory location.
- (==) compares content
- (is) compares memory reference

5. Evaluate the suitability of lists, sets, and dictionaries for string and managing unique value. Justify which data structure is most efficient and why?

→ List: Allows duplicates, slower searching.
Set: Stores only unique value, no indexing.
Dictionaries: Stores unique student IDs as keys and details as value. Most efficient - Dictionary, because it allows fast access, unique keys and structured data storage.

Design and implement a python program that uses:
- Appropriate sequence type
- Indexing or slicing
- Built-in function.

to manage and process student data efficiently

```
Students = {101: "Amit", 102: "Neha", 103: "Rahul"}
print("Student 101:", Students[101])
print("Total students:", len(students))
print("Student IDs:", list(students.key()))
```

Output:

Student 101: Amit

Total students: 3

Student IDs: [101, 102, 103]

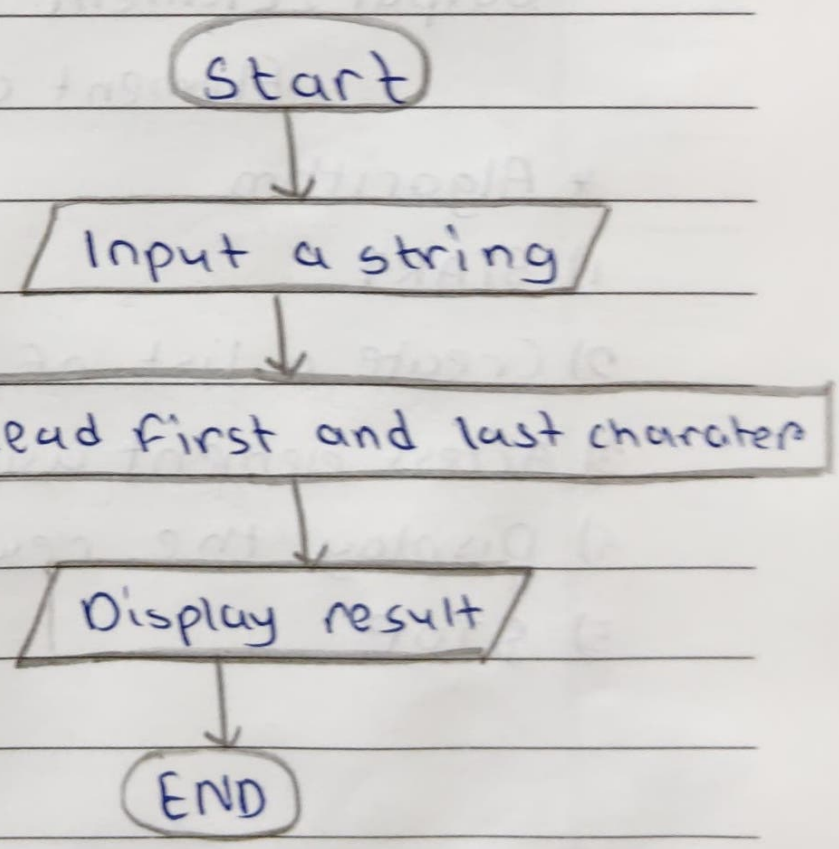
7. Write a program to print the first and last character of a string?

→ s = input("Enter a string:")

```
print("First character:", s[0])
print("Last character:", s[-1])
```

Output: First character: P

Last character: n

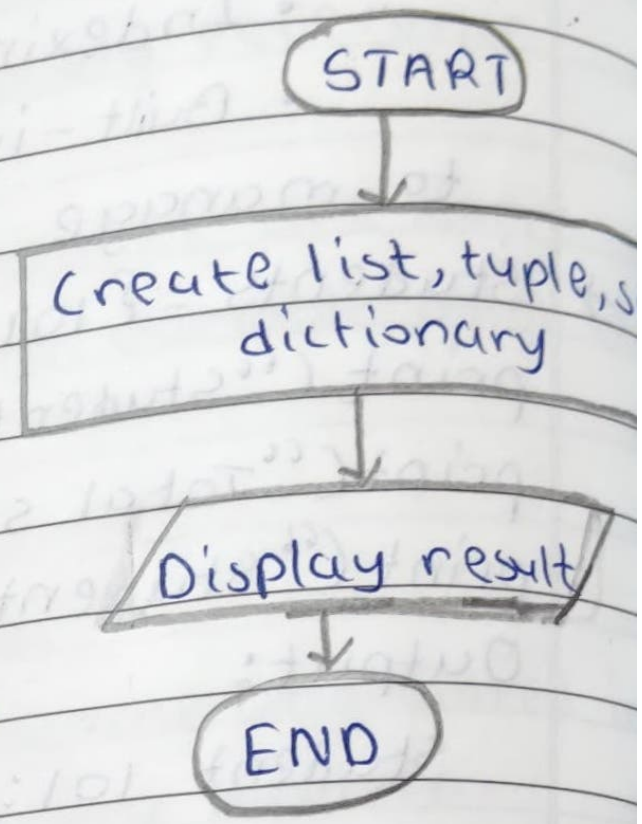


* Algorithm

1) START
2) Read a string
3) Find the first charater
4) Find the last character
5) Display result
6) STOP

8) Write a program to create a list, tuple, set & dict

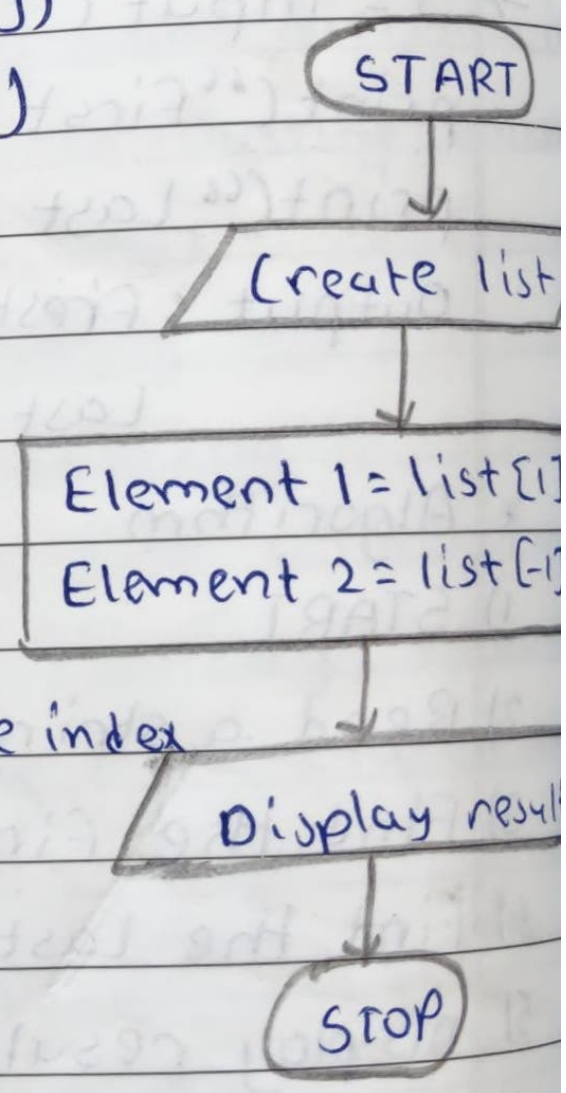
→ list = [1,2,3]
tuple = (4,5,6)
String = {7, 8, 9}
dictionary = {1:"One", 2:"Two"}
Output: [1,2,3]
(4,5,6)



* Algorithm
1) START
2) Create a list, tuple, set, dictionary
3) Display result
4) STOP

9) Write a program to understand the concept of indexin

→ array = [10, 20, 30, 40]
print("Element at index1:" array[1])
print("Element at index-1:" array[-1])
Output: Element at index 1: 20
Element at index -1: 40

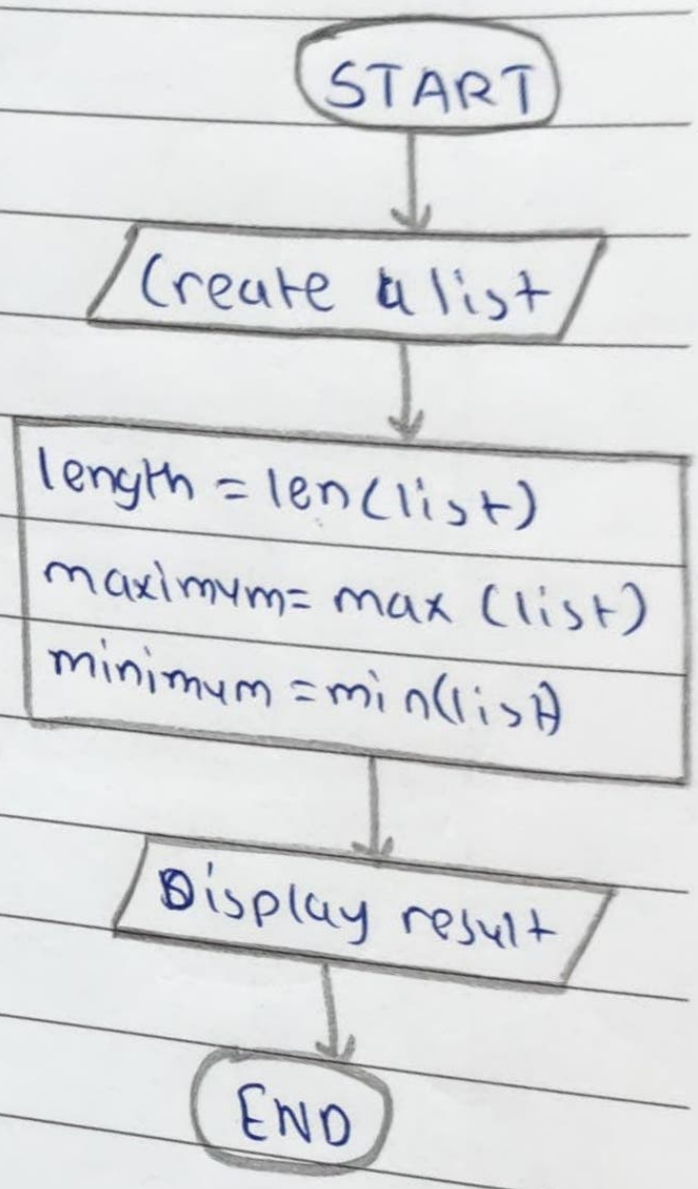


* Algorithm
1) START
2) Create a list of element
3) Access element using positive and negative index
4) Display the result.
5) STOP

Write a program to understand the concept of built-in functions?

```
num[5,2,9,1]
print ("length:", len(num))
print ("Maximum:", max(num))
print ("Minimum:", min(num))
print ("sorted list:", sorted(num))
```

Output: length: 4
Maximum: 9
Minimum: 1
Sorted list: [1,2,5,9]



* Algorithm:

1 Start
2 Create list
3 Use len() function, max() and min() function
4 END